IntelliTwin: Decision-Aware, Condition-Adaptive Conformal Calibration for Turbofan Remaining Useful Life Prognostics

Research synopsis — working draft; final results pending

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Abstract—This research synopsis outlines the theoretical motivation, technical architecture, and confirmatory evaluation framework of the IntelliTwin initiative. The study investigates whether decision-aware, condition-adaptive conformal calibration can improve sequential maintenance utility while preserving reliable prediction-interval coverage for turbofan engine remaining useful life (RUL) estimation. We present the frozen experimental methodology, multi-regime dataset partitioning, base predictor matrix, exact rational engine-weighted quantiles, and 10,000-draw cluster-bootstrap evaluation protocol locked under Architecture Decision Record ADR-018. Confirmatory empirical results on held-out test data remain pending experimental execution.

Index Terms—Predictive maintenance, remaining useful life, conformal prediction, hierarchical inference, decision-aware calibration, digital twins, turbofan engines.

I. Introduction

Prognostics and Health Management (PHM) for high-value rotating machinery, such as commercial aviation turbofan engines, is critical to modern industrial operations [1], [2]. Operational maintenance strategies have traditionally navigated a severe trade-off between corrective maintenance and time-based preventive overhauls. Run-to-failure corrective strategies maximize the physical utilization of machine life but risk uncontained in-service failures, introducing severe flight delays and catastrophic safety hazards. Conversely, scheduled overhauls at fixed flight intervals prevent unexpected shutdowns but scrap substantial components with significant Remaining Useful Life (RUL) intact [3].

Predictive Maintenance (PdM) leverages multi-sensor telemetry to estimate RUL dynamically. Although modern deep neural networks and gradient boosting regressors achieve low point-prediction errors, point estimates $\hat{y}$ provide no measure of predictive uncertainty. In sequential replacement decisions, treating deterministic point estimates as ground truth is operationally hazardous: overestimating remaining life leads to in-service breakdown, while underestimating remaining life precipitates premature replacement [4], [5].

To enable mathematically reliable decision boundaries, prognostic models must produce calibrated prediction intervals $C(x) = [L(x), U(x)]$ that guarantee nominal coverage $1-\alpha$. Conformal prediction [6] provides a distribution-free framework ensuring finite-sample marginal coverage under data exchangeability. However, industrial degradation telemetry violates exchangeability due to intra-engine temporal autocorrelation and varied operational regimes. Furthermore, downstream replacement decisions depend primarily on the lower prediction bound $L(x)$, creating an asymmetric operational penalty that symmetric calibration methods fail to address [7], [8].

The IntelliTwin project investigates decision-aware, condition-adaptive conformal calibration on turbofan degradation benchmarks, seeking to align conditional prediction-interval reliability with sequential maintenance utility.

II. Research Problem and Objectives

A. Research Problem Formulation

Existing literature separately evaluates conformal uncertainty calibration, operating-condition diagnostics, and uncertainty-aware maintenance scheduling. What remains insufficiently established is a Remaining Useful Life (RUL) prediction methodology that jointly adapts uncertainty calibration to operating conditions and downstream sequential replacement consequences while preserving reliable interval coverage.

Our study investigates whether decision-aware, condition-adaptive conformal calibration can improve sequential maintenance utility without sacrificing subgroup reliability or merely becoming more conservative.

B. Formal Research Questions

In accordance with the frozen research protocol (ADR-018), four pre-registered questions are evaluated:

- RQ1 (Sequential Maintenance Utility & Coverage): Can the proposed calibration method improve sequential maintenance utility by at least 0.20 SCU

per engine over both global and standard condition-aware calibration while maintaining absolute coverage errors below 0.05 on C-MAPSS FD002?
- RQ2 (Conditional Subgroup Reliability): Can the proposed method improve worst-group coverage error by at least 0.01 across operating regimes and degradation stages without worsening mean prediction interval width or Winkler score by 1.0 cycle or more?
- RQ3 (Decision-Value Attribution): Are observed maintenance utility gains attributable to decision-aware calibration rather than trivial over-conservatism, demonstrated by outperforming four prespecified matched-conservatism control configurations across three families?
- RQ4 (Robustness & Generalization): Does the proposed method retain its maintenance and calibration benefits across an evaluation matrix of six mandatory cells (FD001, FD002, FD004 × GBR, LSTM) without material harm in any individual cell?

III. Related Work and Positioning

A. Prognostic Models and Asymmetric Penalties

Prognostic modeling for aero-engines widely benchmarked on C-MAPSS [1] encompasses deep architectures including recurrent neural networks and gradient-boosted decision trees. While the C-MAPSS scoring metric penalizes overestimation ($d > 0$) more severely than underestimation ($d < 0$), base models are typically trained under symmetric loss functions.

B. Conformal Uncertainty Quantification

Conformal prediction [6] provides distribution-free marginal guarantees. Conformalized Quantile Regression (CQR) has been applied to aero-engines by Diao et al. [5] and Robinson [8]. However, standard split conformal methods assume observation exchangeability. In degradation trajectories, observations are clustered within engines; Lee, Barber, and Willett [9] established hierarchical conformal inference, bounding finite probability mass by $\frac{m}{m+1}$ and balancing mass equally across groups to ensure valid coverage on unseen clusters.

C. Subgroup Diagnostics and Maintenance Utility

Auditing localized reliability under operating-regime shifts [10] and lifecycle degradation stages [11] demonstrates that marginal coverage frequently conceals localized failure in high-stress regimes. Concurrently, utility-directed conformal prediction [7] demonstrated the benefits of decision-aware nonconformity scores. IntelliTwin unites hierarchical engine-weighted conformal inference with asymmetric decision-aware score functions evaluated in sequential maintenance simulation.

IV. Proposed Methodology

A. Causal Sliding Windows and Dataset Partitioning

Evaluations span C-MAPSS FD001 (100 engines), FD002 (260 engines), and FD004 (249 engines). Splits are constructed strictly at the engine level: approx. 50% training, 20% calibration, 10% validation, and 20% held-out testing (Table I).

Sensor preprocessing (17 features: 3 operational settings and 14 non-constant sensors) is fit exclusively on training data. Causal sliding windows of length $W = 30$ cycles with stride 1 and strictly no padding are extracted. Point predictors consume identical chronological information: sequential tensors of shape $(30, 17)$ for LSTM and flattened vectors of shape $(510,)$ for GBR. Training targets are capped at 125 cycles ($y_{\text{train}} = \min(125, T - t)$), while calibration nonconformity scores and evaluation metrics are computed strictly against true uncapped remaining life $y_{\text{true}} = T - t$.

B. Exact Engine-Weighted Conformal Calibration

To account for unequal engine trajectory lengths, each endpoint (i, t) within calibration engine i receives mass $w_{it} = \frac{1}{m \cdot n_i}$, where m is the number of contributing calibration engines. Following Lee et al. [9], the hierarchical empirical distribution bounds finite mass by $\frac{m}{m+1}$, placing remaining mass $\frac{1}{m+1}$ at $+\infty$:

$$F_{\text{hierarchical}}(q) = \frac{1}{m+1} \sum_{i=1}^{m} \frac{1}{n_i} \sum_{t=1}^{n_i} \mathbb{I}(s_{it} \leq q) \qquad (1)$$

Thresholds q^* are computed via exact rational arithmetic using fractions. If an operating regime contains fewer than $m_g = 10$ calibration engines, execution falls back automatically to the global threshold.

C. Decision-Aware Asymmetric Nonconformity Score

The proposed method (decision_aware_hcp) modulates scaling and asymmetry as a function of point prediction p:

$$\sigma(p) = \left(1 + \frac{p}{125}\right)^{\gamma}, \quad z(p) = \exp\left(-\frac{p}{20}\right) \qquad (2)$$

$$a(p) = \sigma(p) \exp(\beta z(p)), \quad b(p) = \sigma(p) \exp(-\beta z(p)) \qquad (3)$$

where $\gamma \in \{0, 1, 2\}$ and $\beta = \beta_{\text{code}} \ln(2)$ with $\beta_{\text{code}} \in \{-1, 0, 1\}$. The nonconformity score and inverted prediction interval are:

$$\text{score}(p, y) = \max\left(\frac{p - y}{a(p)}, \frac{y - p}{b(p)}, 0\right) \qquad (4)$$

$$L = \max\left(0,\ p - q_g \cdot a(p)\right), \quad U = p + q_g \cdot b(p) \qquad (5)$$

At fixed p, q, γ, negative β decreases $a(p)$ by half and doubles $b(p)$ as $p \to 0$, reducing the subtracted lower-side margin. In validation candidate selection, configuration gamma2_betam1 was frozen across all 6 cells under ADR-021.

D. Sequential Maintenance Simulator

A discrete-event simulation engine replays engine cycles from $t = 30$ to $T - 1$, executing the pre-registered policy:

$$\text{Action}(t) = \begin{cases} \text{MAINTAIN_NOW}, & \text{if } L(t) \leq 1.0 \\ \text{CONTINUE}, & \text{otherwise} \end{cases} \quad (6)$$

Under the primary_reference scenario, preventive replacement incurs $C_{\text{PM}} = 1.0$ SCU, unplanned in-service failure at cycle T incurs $C_{\text{FAIL}} = 5.0$ SCU, and wasted life incurs $C_{\text{WASTE}} = 0.20$ SCU per unused cycle.

V. Experimental Evaluation Protocol

A. Statistical Uncertainty and Engine-Level Resampling

To account for trajectory autocorrelation, all comparative inferences use paired engine-cluster bootstrap resampling with replacement ($B = 10{,}000$ resamples) stratified by lifetime quartiles (Q_1–Q_4). The pseudo-random number generator is PCG64 seeded via SeedSequence([2026, 6, dataset_number]).

Multiplicity control across the four research questions is enforced at $\alpha_{\text{fam}} = 0.05/4 = 0.0125$ via Bonferroni correction, yielding **98.75% decision intervals** ($\alpha = [0.00625, 0.99375]$). In accordance with intersection-union testing (IUT), full support for an RQ requires every single component condition to clear its boundary at the 98.75% decision level.

B. Frozen Decision Rules and Practical Margins

- RQ1 (Utility & Coverage): Lower CI of $\Delta_{\text{cost}} > 0.20$ SCU per engine vs. both global_hcp and regime_hcp; upper CI of marginal, worst-regime, and worst-stage coverage errors < 0.05; observed fleet failure rate increase $\Delta_{\text{fail}} \leq 0.0$.
- RQ2 (Subgroup Reliability): Lower CI of $\Delta_{\text{error}} > 0.01$ vs. both baselines across operating regimes (C_1–C_6) and lifecycle stages (Early/Mid/Late); upper CI of worst-group error < 0.05; upper CI of MPIW and interval score expansion $D_{\text{width}}, D_{\text{score}} < 1.0$ cycle.
- RQ3 (Decision-Value Attribution): Lower CI of $\Delta_{\text{cost}} > 0.20$ SCU over all four prespecified matched-conservatism controls across three families (matched width, widened interval, point safety margin, earlier intervention) selected on validation engines under ADR-023.
- RQ4 (Generalization): Weighted aggregate lower CI of $\Delta_{\text{cost}} > 0.20$ SCU and $\Delta_{\text{error}} > 0.01$, combined with cell-level non-harm bounds ($\Delta_{\text{cost}} > -0.20$ SCU, $\Delta_{\text{error}} > -0.01$) across all six mandatory evaluation cells.

VI. Current Project Status and Preliminary Results

A. Completed Preparatory Engineering & Governance

The research preparation phases are complete and immutably frozen:

TABLE I
Dataset Partition Summary (ADR-011)

Dataset	Train (50%)	Cal (20%)	Val (10%)	Test (20%)
FD001	50	20	10	20
FD002	130	52	26	52
FD004	124	50	25	50
Total	304	122	61	122

- Phase 0–3 (Governance & Splits): Packaging verified under ADR-001; C-MAPSS dataset splits frozen across 609 engines in data/splits/; 17-feature schema locked under ADR-013.
- Phase 5 (Base Predictors): Complete 30-model matrix trained across five official seeds (2026–2030) under ADR-019 and ADR-020; all 30 checkpoints verified on CPU in experiments/base_models/model_registry.json.
- Phase 6 (Conformal Calibration): 90 calibrator bundles fitted and verified under ADR-021; configuration gamma2_betam1 frozen across all six cells in experiments/calibration/selection/selection_freeze.json.
- Phase 7 Pre-Test Preparation: Pre-test evaluation machinery, 10,000-draw cluster bootstrap engine, and four matched controls frozen under ADR-023 in experiments/confirmatory_development/protocol_freeze.json.

B. Honest Demarcation: Results Strictly Pending

In accordance with preregistration ethics:

1) Held-Out Confirmatory Evaluation: Strictly NOT STARTED. Zero held-out test windows have been evaluated (pretest_audit.json).
2) RQ1–RQ4 Outcomes: Strictly PENDING. No performance numbers or hypothesis outcomes are reported from held-out data in this draft.
3) Pending Corrective Audit: Official Step 8 checkpoint prediction reproduction was recorded as NOT RUN in step08_repair_verification.json; formal reverification is pending under experiments/pretest_verification_v2/ (UPD-01).

VII. Limitations, Threats to Validity, and Conclusion

A. Research Limitations

1) Benchmark Simulation: Evaluations are conducted on synthetic NASA C-MAPSS data where degradation is monotonic. Operational aircraft engines experience maintenance reversibility and non-engine sensor faults not captured in the benchmark.
2) Synthetic Economic Scenarios: The Simulated Cost Units ($C_{\text{PM}} = 1.0, C_{\text{FAIL}} = 5.0, C_{\text{WASTE}} = 0.20$) represent study-defined benchmark preferences, not empirical airline accounting constants.
3) Conformal Theory Boundaries: Conformal calibration guarantees marginal exchangeable coverage.

It does not provide theoretical guarantees of exact pointwise conditional coverage or sequential stopping-time coverage.

4) No Physical Integration: The framework evaluates an offline, trace-driven simulation replay engine; no physical engine hardware coupling exists.

B. Conclusion and Next Milestones

IntelliTwin establishes an end-to-end, scientifically reproducible research architecture linking prognostic neural networks, hierarchical conformal calibration, and sequential maintenance simulation. All methodology, base models, calibrator bundles, and evaluation contracts are completely frozen and audited. Upon completion of the official held-out test evaluation, verified empirical results will be incorporated to resolve RQ1–RQ4.

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